

Name of the Student	D.N Shirshika
Reg. No	192425396
GitHub Link	https://github.com/shirshika572/MLa0403.git

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 1

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Duration: 90 minutes

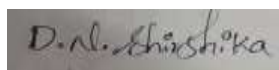
Assessment Tool Weightage: 30%

Mark : 25

Code of Conduct:

I D.N Shirshika 192425396 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: D.N Shirshika

Criteria	Problem Understanding & Requirement Analysis (2)	Model Architecture & Design (2.5)	Application of Deep Learning Techniques (2.5)	Model Development & Implementation Strategy (2)	Performance Analysis & Justification (1)	Total (10)
Weightage	20 %	30%	25%	15 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

QUESTIONS:

05

Total Marks:25

1. **Design and develop a model for a hospital** has only **2,000 labeled X-ray images** for classifying images into normal and abnormal categories. A pre-trained ResNet model is available. **Design a CNN-based transfer learning model** by identifying which layers should be frozen, which layer should be replaced, and how the model should be trained

Introduction

A hospital has only 2,000 labeled X-ray images for classifying medical images into **normal and abnormal** categories. Training a deep CNN from the beginning requires a large amount of labeled data and computational resources. Therefore, **transfer learning** can be used with a pre-trained ResNet model. A ResNet model trained on a large image dataset can provide useful low-level and high-level image features that can be adapted to X-ray classification.

Proposed Model

A **pre-trained ResNet-50** can be used as the base CNN model.

Architecture

X-Ray Image



Image Preprocessing



Pre-trained ResNet-50



Freeze Initial Convolutional Layers



Fine-tune Last ResNet Block



Global Average Pooling



Fully Connected Layer



Dropout



Sigmoid Output



Normal / Abnormal

Layers to Freeze

The initial convolutional layers of ResNet learn general features such as:

- Edges
- Shapes
- Textures
- Basic patterns

These features are useful for X-ray images, so the **initial layers should be frozen**.

For example:

- Freeze the initial convolutional layers.
- Freeze most of the middle ResNet blocks.
- Optionally fine-tune the final ResNet block because it contains more task-specific features.

Thus, only the final portion of the network needs to be trained on the X-ray dataset.

Training Process

First, the 2,000 X-ray images are divided into training, validation, and testing sets. Images are resized to the required ResNet input size and normalized.

Data augmentation such as:

- Rotation
- Horizontal flipping where medically appropriate
- Zooming
- Small translations

can be applied to increase the effective variety of training data.

Initially, the frozen ResNet feature extractor is kept unchanged and only the new classification layer is trained. After the classifier becomes stable, the final ResNet block can be unfrozen and trained with a **small learning rate**.

Advantages

- Requires less training data.
- Reduces training time.
- Uses already learned visual features.
- Reduces the risk of overfitting.
- Provides better performance than training a CNN from scratch in many small-data situations.

Conclusion

A pre-trained ResNet model with frozen early layers and a newly trained binary classification head is suitable for the hospital's X-ray classification problem. Fine-tuning only the final ResNet layers allows the model to adapt general image features to medical X-ray patterns while reducing overfitting.

2. An agricultural organization wants to identify **five types of plant diseases** from leaf images. Since the dataset contains limited training samples, the organization wants to reuse features efficiently. **Design a DenseNet-based classification model** and explain how dense connections can help the model reuse features during classification.

Introduction

An agricultural organization wants to classify leaf images into **five different plant disease categories**. Since the available training dataset contains limited samples, a DenseNet architecture is suitable because it promotes **feature reuse** through dense connections between layers.

Proposed Model

A pre-trained **DenseNet-121** can be used as the feature extraction network.

Architecture

Leaf Image



Image Preprocessing



Pre-trained DenseNet-121



Dense Blocks



Transition Layers



Global Average Pooling



Fully Connected Layer



Softmax



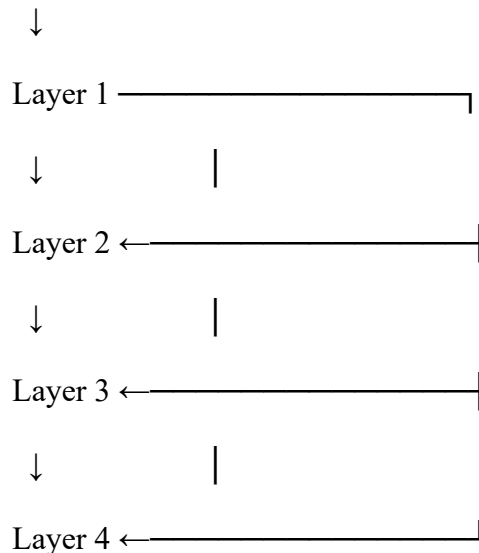
5 Plant Disease Classes

Dense Connections

In a traditional CNN, a layer generally receives input only from the previous layer.

DenseNet works differently. Each layer receives the feature maps from **all preceding layers within a dense block**.

Input



The outputs of earlier layers are directly available to later layers.

Feature Reuse

Early layers may learn:

- Edges
- Colors
- Simple textures

Middle layers may learn:

- Leaf shapes
- Veins
- Spots
- Patterns

Later layers can combine these features to recognize diseases.

Dense connections allow later layers to directly access earlier features instead of relearning them.

Classification Layer

The original DenseNet classification layer can be replaced with a new classification head:

DenseNet Features



Global Average Pooling



Dropout



Dense Layer



Softmax



5 Classes

Training

Initially, the pre-trained DenseNet layers can be frozen and the new classification layer can be trained. After that, selected upper DenseNet layers can be unfrozen and fine-tuned using a small learning rate.

Image augmentation can be used to increase training diversity.

Advantages

- Efficient feature reuse.
- Better information flow between layers.
- Helps reduce redundant feature learning.
- Requires fewer parameters than some conventional architectures.
- Useful for limited datasets.
- Can capture fine details such as leaf spots and texture changes.

Conclusion

DenseNet is well suited for plant disease classification because dense connections allow the network to reuse features from earlier layers. This is particularly useful when the training dataset is limited because the model can make effective use of previously learned visual representations.

3. A university wants to predict a student's future academic performance using their **weekly attendance, assignment marks, internal assessment marks, and previous examination scores** collected over several weeks. **Design an LSTM-based model** that can process this sequential information and predict the student's performance category.

Introduction

Student academic performance depends on information collected over time. Weekly attendance, assignment marks, internal assessment marks, and previous examination scores form a **sequence of data**. An LSTM network is suitable because it can learn patterns and dependencies from sequential data.

Input Data

For every week, the model receives four features:

Attendance

Assignment Marks

Internal Assessment Marks

Previous Examination Score

For example:

Week 1 → [Attendance, Assignment, Internal, Exam]

Week 2 → [Attendance, Assignment, Internal, Exam]

Week 3 → [Attendance, Assignment, Internal, Exam]

...

Proposed Architecture

Weekly Student Data



Data Preprocessing



Normalization



Sequence Formation



LSTM Layer



Dropout



Dense Layer



Softmax



Performance Category

LSTM Working

An LSTM contains memory mechanisms that help it retain important information from previous time steps.

The main components are:

- Forget gate
- Input gate
- Cell state
- Output gate

The forget gate decides which previous information should be discarded. The input gate determines which new information should be stored. The cell state carries important information through the sequence.

Therefore, the LSTM can identify patterns such as:

Consistently low attendance and declining assignment marks may indicate poor future performance.

Model Design

Suppose data from 8 weeks is used.

The input shape can be:

8 time steps $\times$ 4 features

The model can contain:

Input



LSTM Layer



Dropout



Dense Layer



Softmax

The final layer can have three neurons if the categories are:

Poor

Average

Excellent

Training

The dataset is divided into training and testing data. The numerical features are normalized before training.

The LSTM learns from previous weekly sequences and predicts the student's future performance category.

Categorical cross-entropy can be used as the loss function for three or more performance categories..

Advantages

- Suitable for sequential data.
- Remembers information from previous weeks.
- Can identify performance trends.
- Handles temporal dependencies.
- Useful for early prediction of academic performance.

Conclusion

An LSTM-based model can effectively process weekly academic information because student performance is a time-dependent problem. The model learns from historical sequences of attendance and marks and predicts the student's future performance category

4. An organization wants to classify customer reviews as **positive, negative, or neutral**. Since the meaning of a word may depend on both the words appearing before and after it, **design a Bidirectional RNN model** for this application. Identify the major processing stages from input text to final sentiment prediction.

Introduction

An organization wants to classify customer reviews into **positive, negative, and neutral** categories. The meaning of a word can depend on both previous and following words. Therefore, a **Bidirectional RNN** is appropriate because it processes the text in both forward and backward directions.

For example:

"The product is not very good."

The word "good" must be interpreted using the preceding words "not very."

Proposed Architecture

Customer Review



Text Cleaning



Tokenization



Word Embedding



Bidirectional RNN



Forward RNN Backward RNN



Combined Representation



Dense Layer



Softmax



Positive / Negative / Neutral

Major Processing Stages

1. Input Text

The customer review is given as raw text.

Example:

"The product quality is excellent."

2. Text Preprocessing

The text can be cleaned by:

- Converting text to a standard case.
- Removing unnecessary symbols.
- Tokenizing the sentence.
- Converting words into numerical representations.

3. Word Embedding

Each word is converted into a numerical vector.

For example:

product → [0.21, 0.43, ...]

quality → [0.15, 0.62, ...]

excellent → [0.72, 0.81, ...]

4. Bidirectional RNN

The sentence is processed in two directions.

Forward RNN:

The → product → quality → is → excellent

Backward RNN:

excellent → is → quality → product → The

The outputs from both directions are combined.

Therefore, the model obtains information from both the preceding and following words.

5. Dense Layer

The combined representation is passed to a fully connected layer that learns the relationship between the extracted text features and sentiment categories.

6. Softmax Output

The final Softmax layer produces three probabilities:

Positive

Negative

Neutral

Example:

Positive → 0.92

Negative → 0.03

Neutral → 0.05

Prediction:

Positive

Advantages

- Uses context from both directions.
- Understands relationships between words.
- Suitable for sentiment analysis.
- Better handles context-dependent words.
- Can capture meaningful sentence representations.

Conclusion

A Bidirectional RNN is suitable for sentiment classification because it considers both previous and subsequent words. The complete review is converted into embeddings, processed in forward and backward directions, and finally classified into positive, negative, or neutral sentiment.

5. A software company wants to develop a system that translates **English sentences into Hindi**. The length of the input and output sentences may differ, and the system needs to understand the complete input sentence before generating the translated sentence. **Design an Encoder–Decoder model** and describe the role of the encoder, decoder, hidden representation, and output layer in the translation process.

Introduction

A software company wants to translate English sentences into Hindi. Since the input and output sentences may have different lengths, a fixed one-to-one mapping between words is not sufficient. An **Encoder–Decoder architecture**, also known as a sequence-to-sequence model, can solve this problem.

Proposed Architecture

English Sentence



Tokenization



Word Embedding



Encoder



Hidden Representation



Decoder



Output Layer



Hindi Sentence

Example

Input:

English: "I am going to school."

Output:

Hindi: "मैं स्कूल जा रहा हूँ।"

The number and arrangement of words can differ between the two languages.

1. Encoder

The encoder receives the English sentence one token at a time.

I → am → going → to → school

The encoder converts the words into numerical representations using an embedding layer and processes them using an RNN, LSTM, or GRU.

Its main purpose is to understand the input sentence and create a meaningful internal representation.

2. Hidden Representation

The encoder produces a hidden representation containing information about the input sentence.

It should capture important information such as:

- Words in the sentence
- Word relationships
- Sentence meaning
- Context

This representation is passed to the decoder.

3. Decoder

The decoder generates the Hindi sentence one token at a time.

.

4. Output Layer

The decoder output is passed through a Softmax layer.

The Softmax layer calculates the probability of possible Hindi words...

.

Teacher forcing can be used during training, where the correct previous target word is supplied to the decoder.

Advantages

- Handles different input and output lengths.
- Learns relationships between words.
- Can be used for machine translation.
- Encoder and decoder can be trained together.
- Can be extended with attention mechanisms to better handle long sentences.

Conclusion

The Encoder–Decoder model is suitable for English-to-Hindi translation because the encoder understands the complete English input and creates a hidden representation, while the decoder uses this representation to generate the Hindi sentence sequentially. The output layer predicts the most appropriate Hindi word at every step.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Requirement Analysis	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Model Architecture & Design	30%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Application of Deep Learning Techniques	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Model Development & Implementation Strategy	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Performance Analysis & Justification	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis